



US012413687B2

(12) **United States Patent**
Lin et al.

(10) **Patent No.:** **US 12,413,687 B2**

(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **PROVIDING SIDE CONVERSATIONS
WITHIN A VIRTUAL CONFERENCING
SYSTEM**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/185,246**

(22) Filed: **Mar. 16, 2023**

(65) **Prior Publication Data**

US 2023/0216991 A1 Jul. 6, 2023

Related U.S. Application Data

(63) Continuation of application No. 17/390,510, filed on
Jul. 30, 2021, now Pat. No. 11,683,447.

(Continued)

(51) **Int. Cl.**
G06F 3/0481 (2022.01)
H04N 7/15 (2006.01)

(52) **U.S. Cl.**
CPC **H04N 7/157** (2013.01); **G06F 3/0481**
(2013.01); **H04N 7/15** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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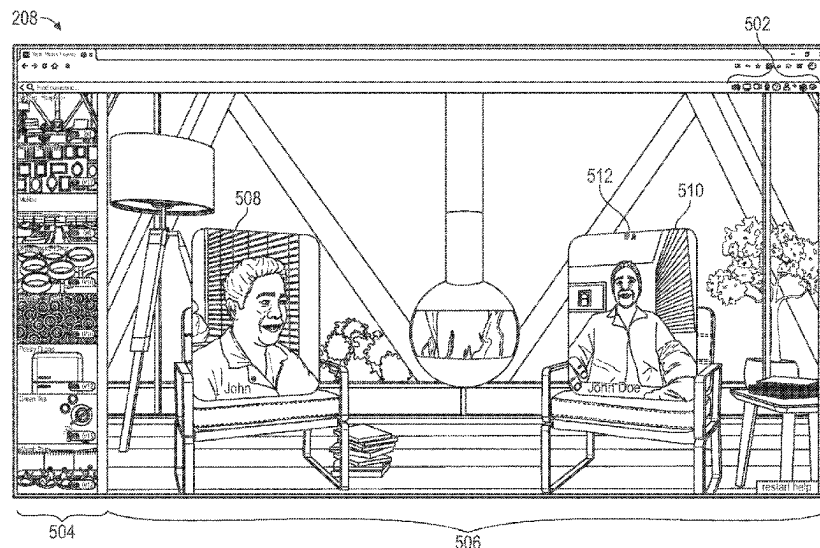
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Woessner, P.A.

(57) **ABSTRACT**

Aspects of the present disclosure involve a system compris-
ing a computer-readable storage medium storing a program
and method for providing side conversations within a virtual
conferencing system. The program and method provide, in
association with a virtual conference among plural partici-
pants, a first audio channel for sharing among the plural parti-
cipants; provide, for each of the plural participants,
display of a participant video element which corresponds to
the participant and which includes a user-selectable button
to initiate a side conversation with the participant; receive
indication of a user selection, by a first participant, of the
user-selectable button for initiating a side conversation with
a second participant of the plural participants; and provide a
second audio channel from a first device associated with the
first participant to a second device associated with the
second participant, in conjunction with providing the first
audio channel to the second device.

20 Claims, 10 Drawing Sheets



Related U.S. Application Data

- (60) Provisional application No. 63/168,033, filed on Mar. 30, 2021.

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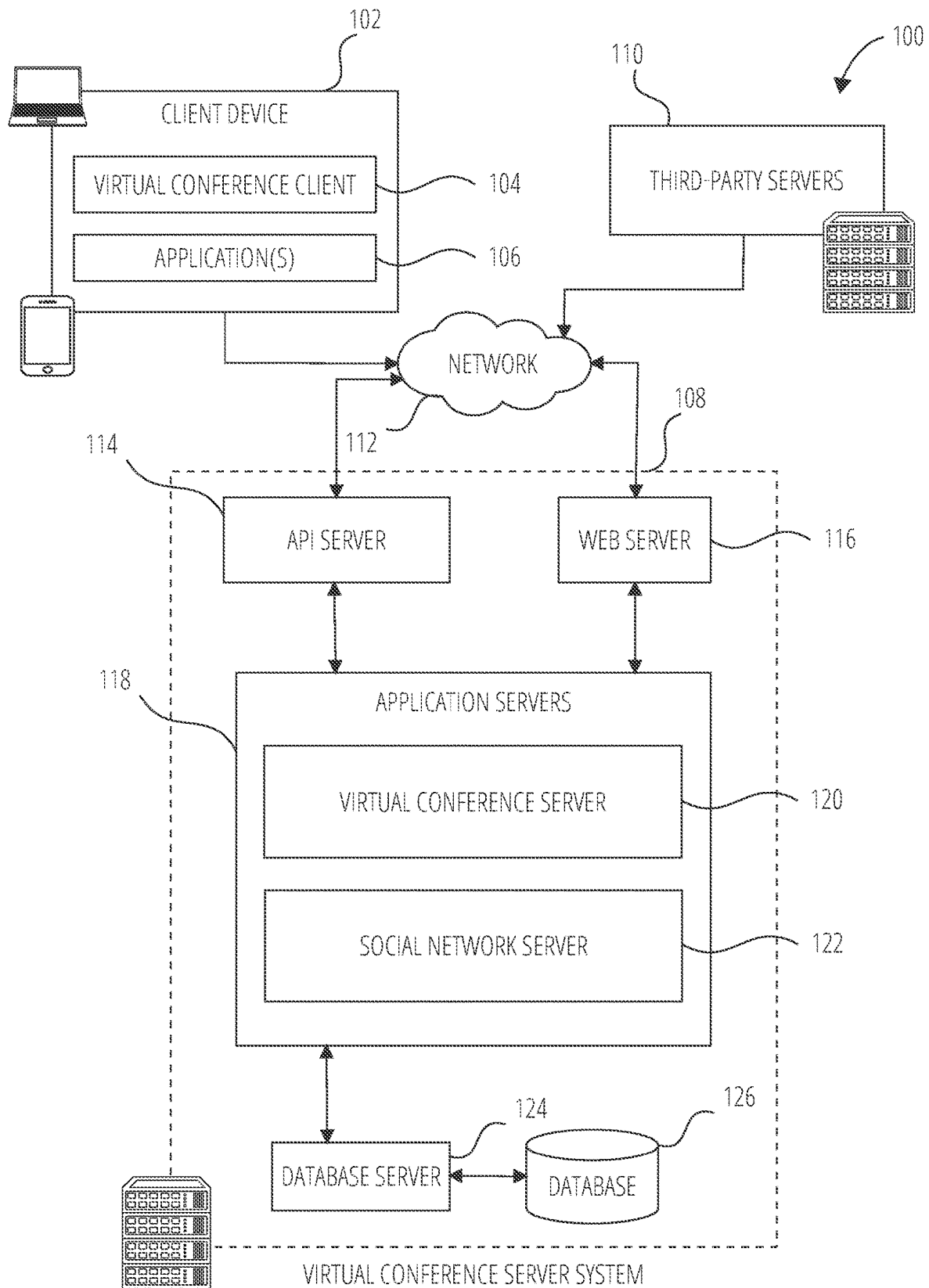


FIG. 1

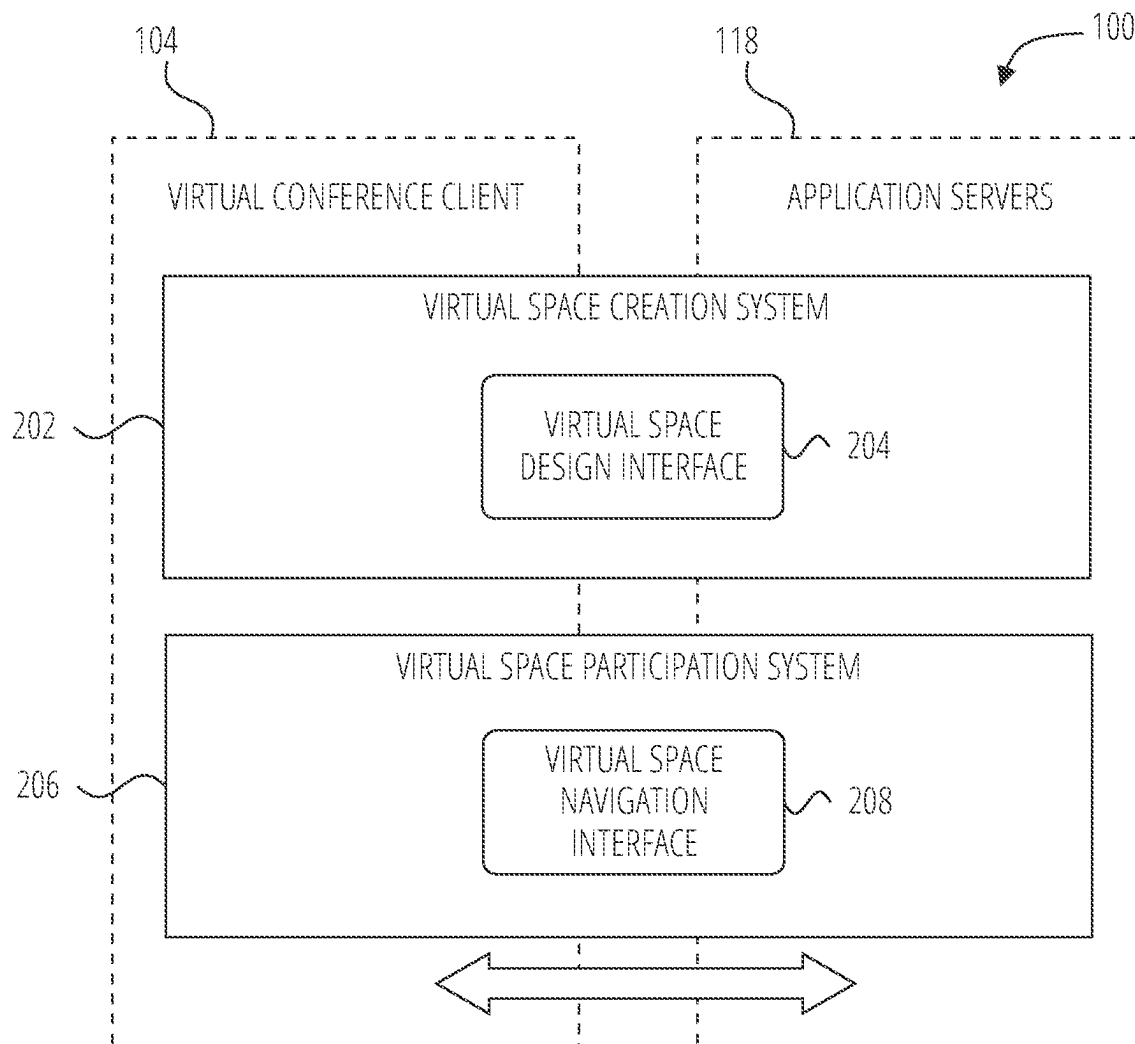
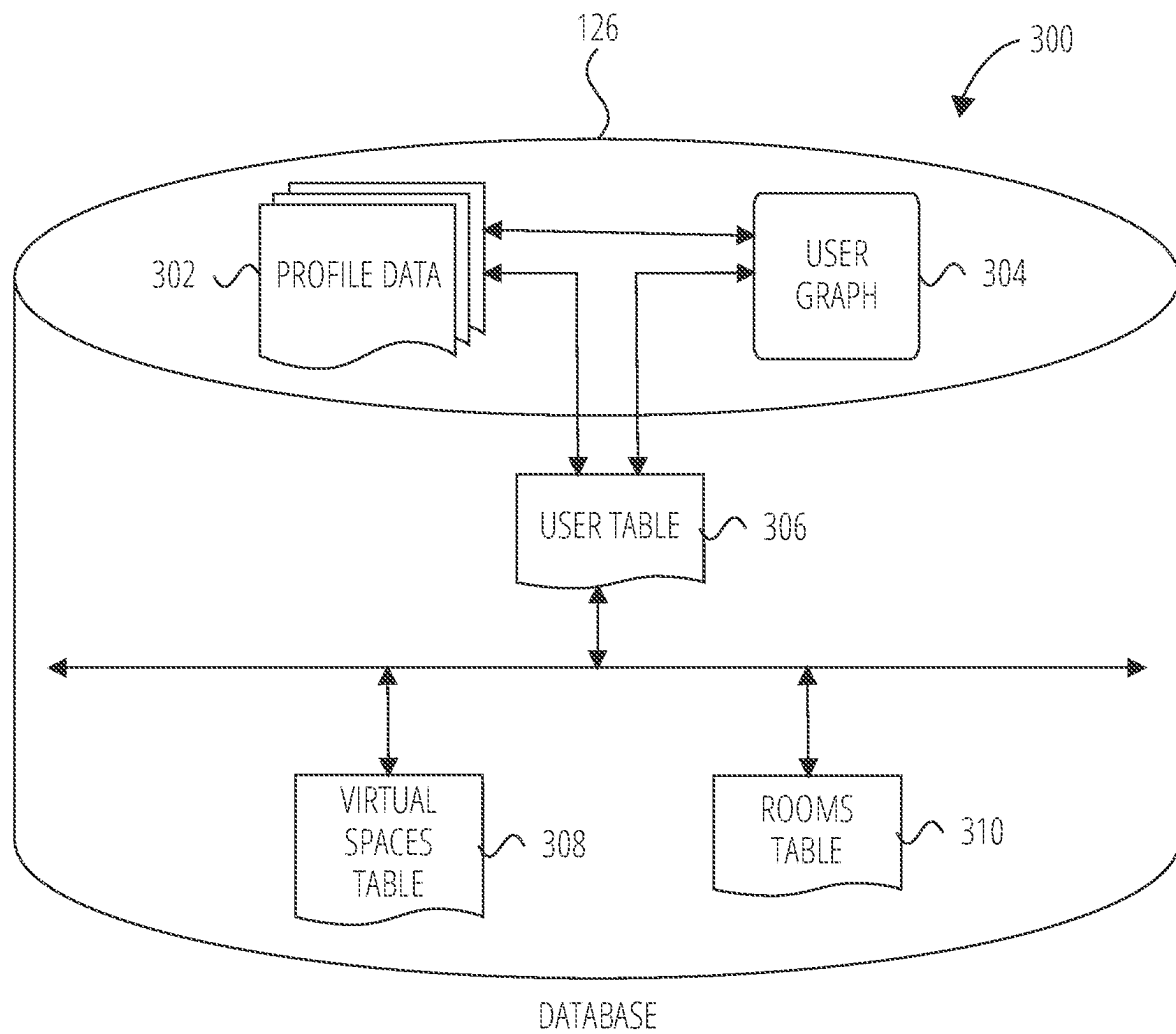
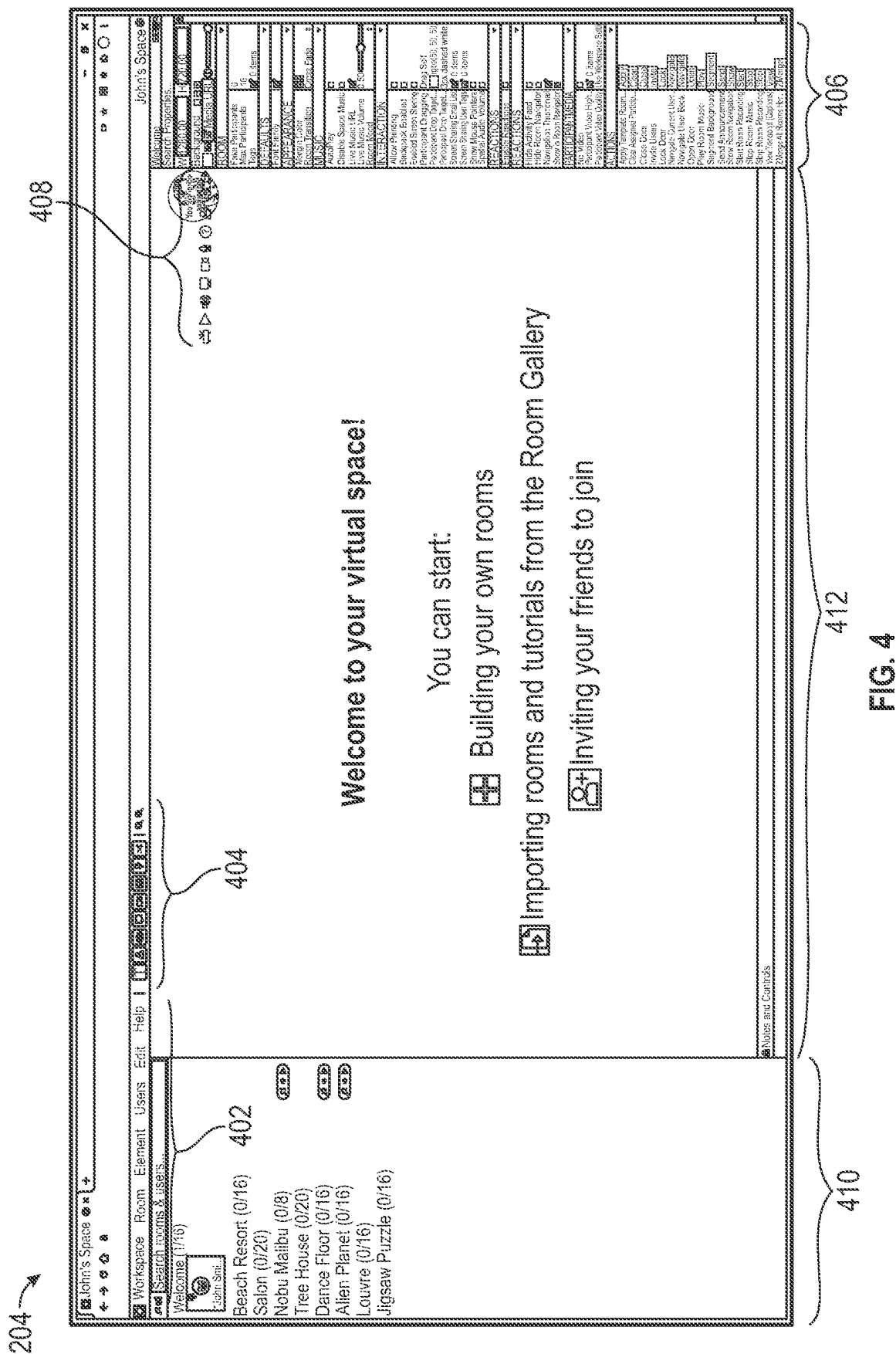
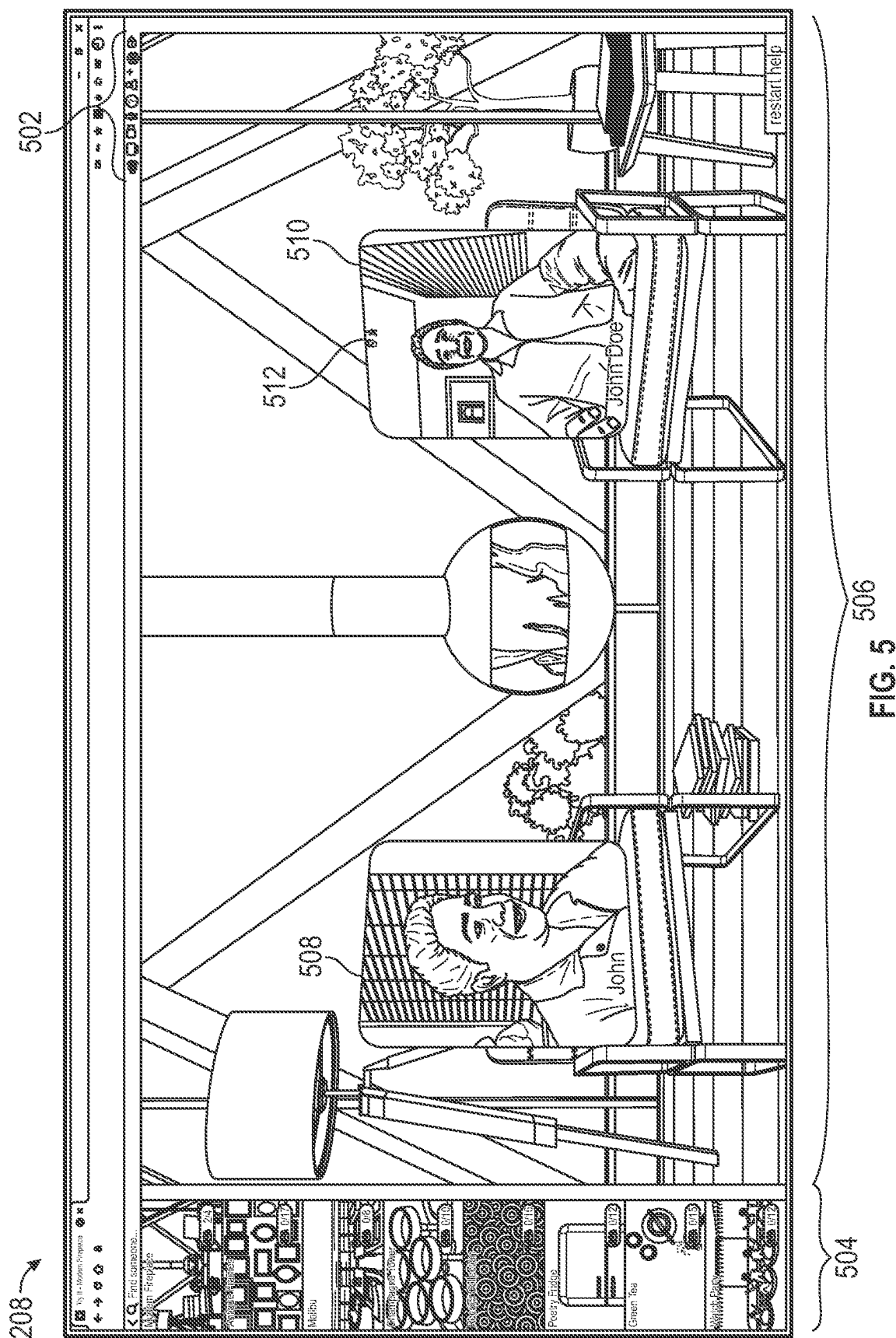
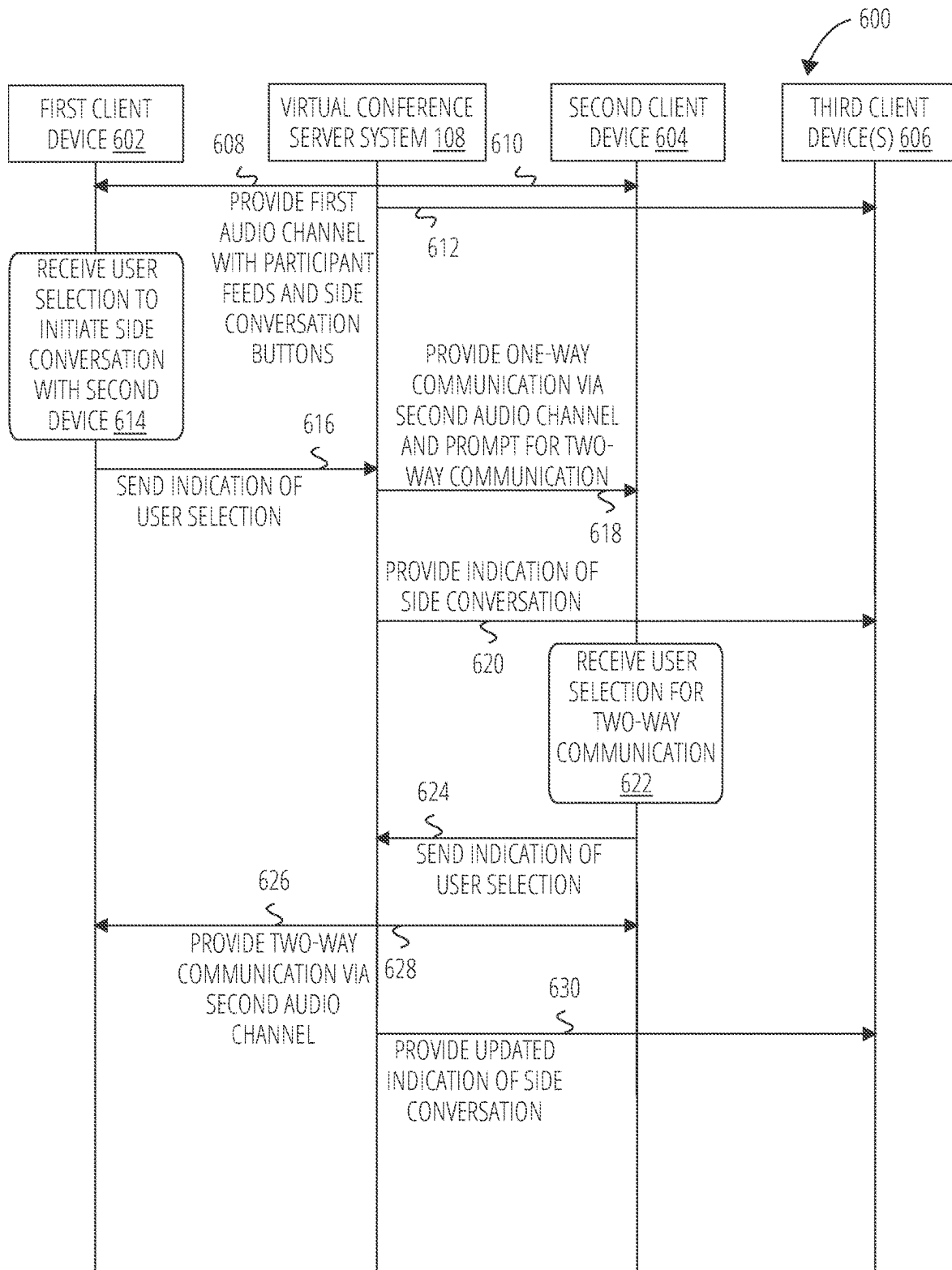


FIG. 2

**FIG. 3**







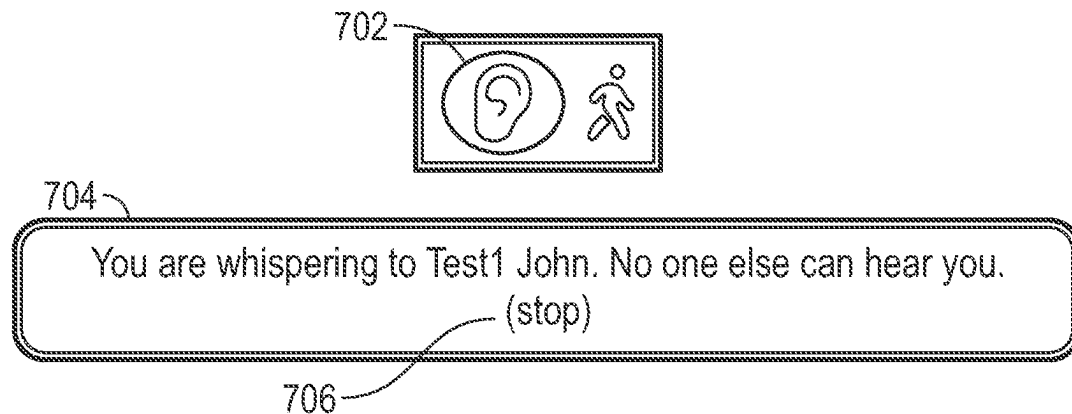


FIG. 7A

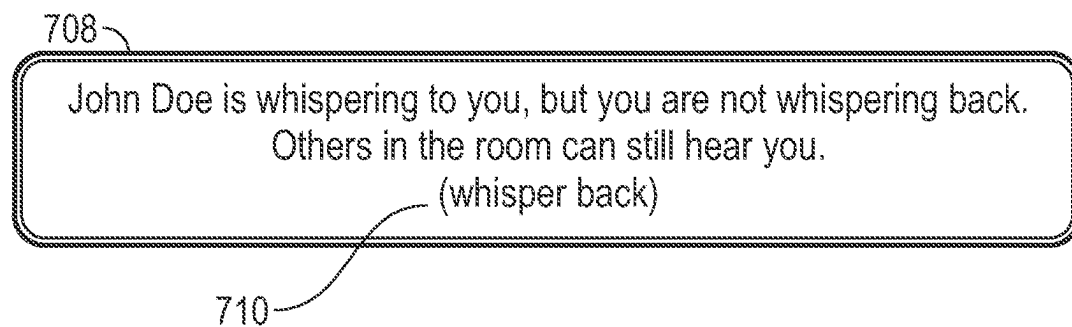


FIG. 7B

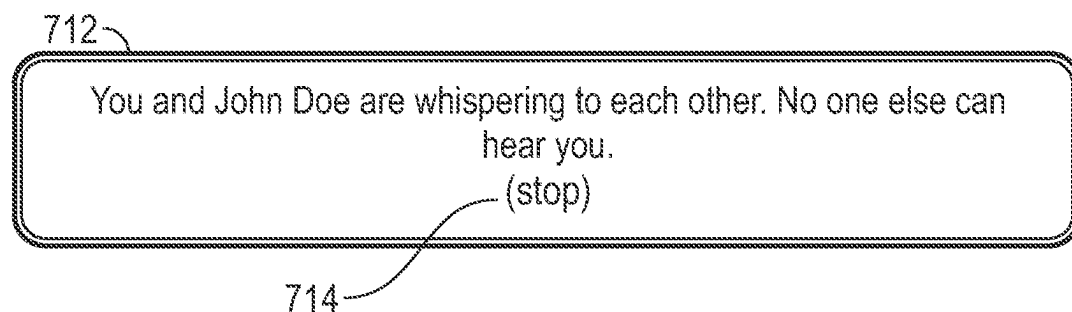
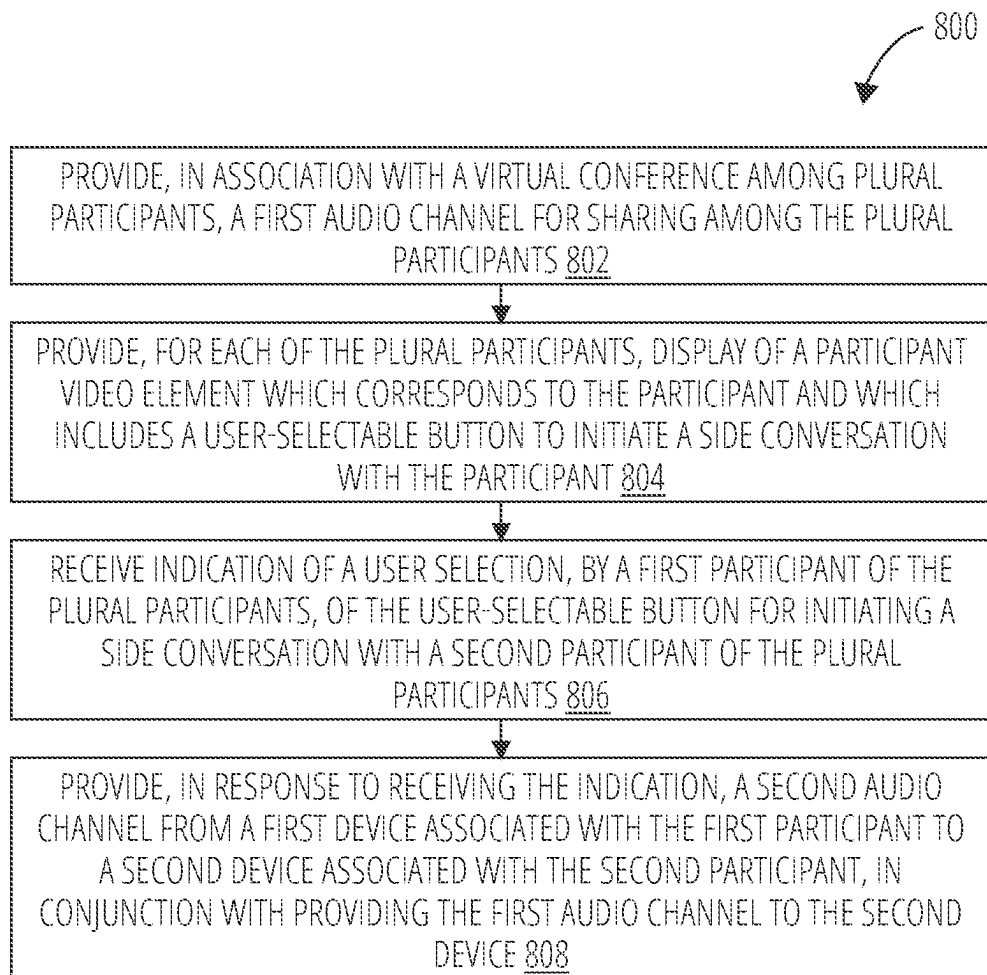


FIG. 7C

**FIG. 8**

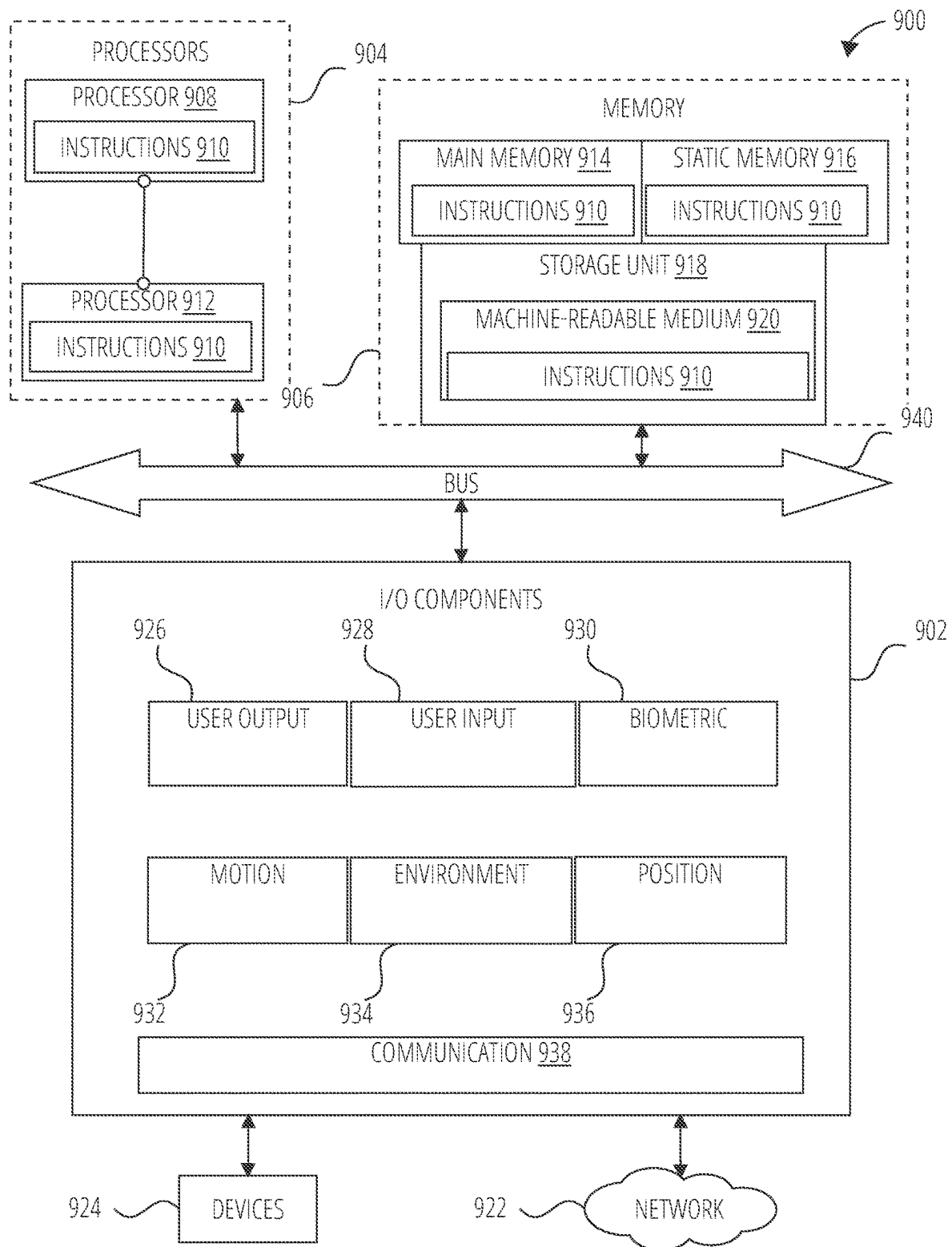


FIG. 9

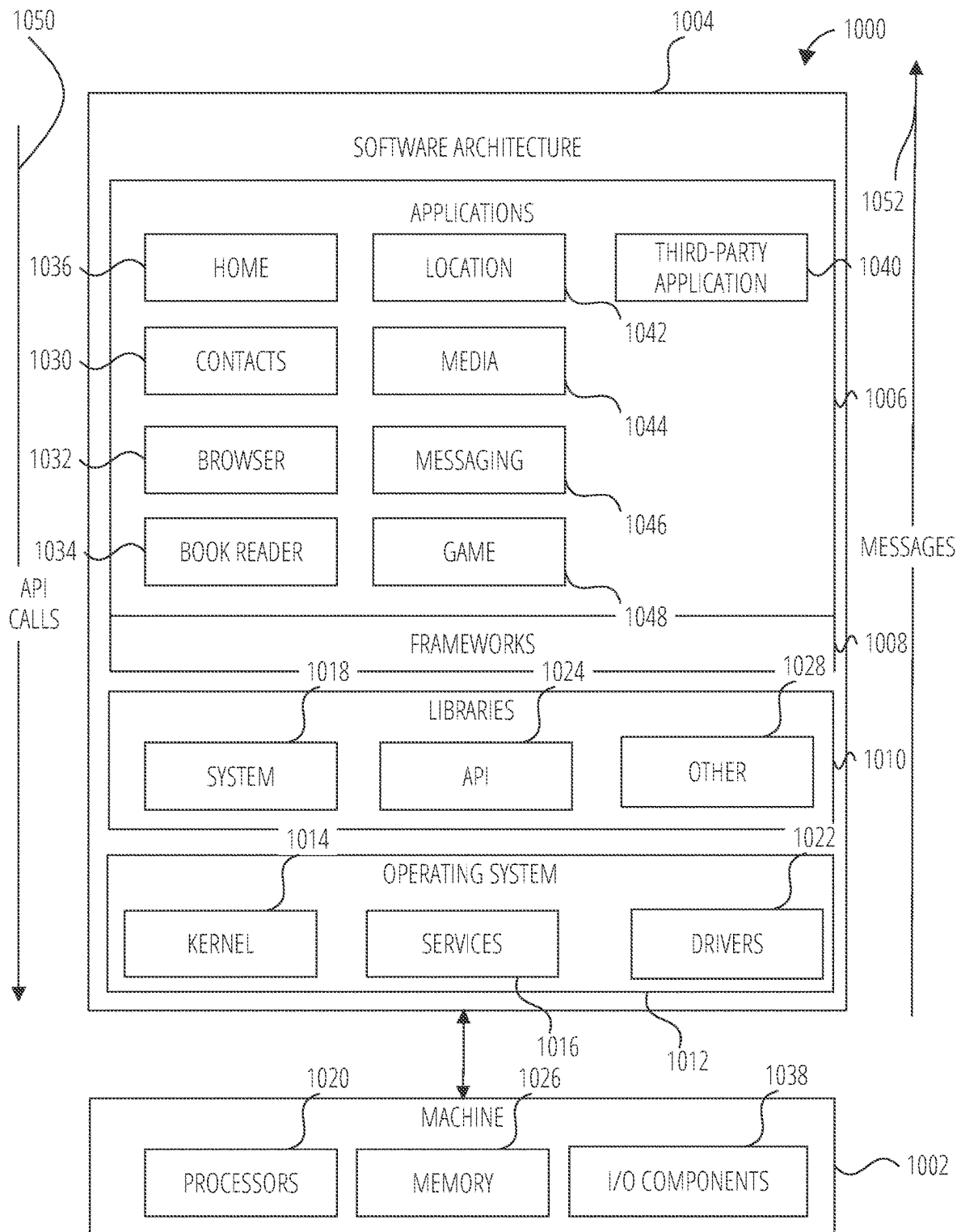


FIG. 10

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PROVIDING SIDE CONVERSATIONS WITHIN A VIRTUAL CONFERENCING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATION

This patent application is a continuation of U.S. patent application Ser. No. 17/390,510, filed Jul. 30, 2021, which application claims the benefit of U.S. Provisional Patent Application No. 63/168,033, filed Mar. 30, 2021, entitled “PROVIDING SIDE CONVERSATIONS WITHIN A VIRTUAL CONFERENCING SYSTEM”, which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

The present disclosure relates generally to virtual conferencing systems, including providing side conversations within a virtual conferencing system.

BACKGROUND

A virtual conferencing system provides for the reception and transmission of audio and video data between devices, for communication between device users in real-time.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some nonlimiting examples are illustrated in the figures of the accompanying drawings in which:

FIG. 1 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

FIG. 2 is a diagrammatic representation of a virtual conferencing system, in accordance with some examples, that has both client-side and server-side functionality.

FIG. 3 is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

FIG. 4 illustrates a virtual space design interface with interface elements for designing a virtual space, in accordance with some example embodiments.

FIG. 5 illustrates a virtual space navigation interface with interface elements to navigate between the rooms of a virtual space and to participate in virtual conferencing with respect to the rooms, in accordance with some example embodiments.

FIG. 6 is an interaction diagram illustrating a process for providing a side conversation within a virtual conferencing system, in accordance with some example embodiments.

FIGS. 7A-7C illustrate various interface elements relating to providing a side conversation between participants of a virtual conference, in accordance with some example embodiments.

FIG. 8 is a flowchart illustrating a process for participating in a side conversation within a virtual conferencing system, in accordance with some example embodiments.

FIG. 9 is a diagrammatic representation of a machine in the form of a computer system within which a set of

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instructions may be executed for causing the machine to perform any one or more of the methodologies discussed herein, in accordance with some examples.

FIG. 10 is a block diagram showing a software architecture within which examples may be implemented.

DETAILED DESCRIPTION

A virtual conferencing system provides for the reception and transmission of audio and video data between devices, for communication between device users in real-time. In some cases, participants (e.g., two participants) within a virtual conference may wish to engage in a side conversation, without disrupting other participants in the virtual conference.

The disclosed embodiments provide for initiating a side conversation between participants of a virtual conference. The virtual conferencing system provides a first audio channel (e.g., a primary audio channel) for the virtual conference among multiple participants. Each of the participants is represented by a respective participant video element which corresponds to that participant. In addition, each participant video element includes a user-selectable button to initiate a side conversation with that participant.

A first participant may select the button to initiate a side conversation with a second participant. The virtual conferencing system initiates the side conversation as a one-way communication, via a second audio channel, from the first participant's device to the second participant's device. During the one-way communication, the audio level of the first audio channel is reduced on the second participant's device, so as to emphasize the audio level of the second audio channel. In addition, the virtual conferencing system prompts the second participant for two-way communication, so that the second participant may communicate back with the first participant via the second audio channel. Thus, the virtual conferencing system provides intuitive, simple interfaces for initiating and engaging in side conversations with respect to virtual conferencing.

FIG. 1 is a block diagram showing an example virtual conferencing system 100 for exchanging data over a network. The virtual conferencing system 100 includes multiple instances of a client device 102, each of which hosts a number of applications, including a virtual conference client 104 and other application(s) 106. Each virtual conference client 104 is communicatively coupled to other instances of the virtual conference client 104 (e.g., hosted on respective other client devices 102), a virtual conference server system 108 and third-party servers 110 via a network 112 (e.g., the Internet). A virtual conference client 104 can also communicate with locally-hosted applications 106 using Applications Program Interfaces (APIs).

The virtual conferencing system 100 provides for the reception and transmission of audio, video, image, text and/or other signals by user devices (e.g., at different locations), for communication between users in real-time. In some cases, two users may utilize virtual conferencing to communicate with each other in one-to-one communication at their respective devices. In other cases, multiway virtual conferencing may be utilized by more than two users to participate in a real-time, group conversation. Thus, multiple client devices 102 may participate in virtual conferencing, for example, with the client devices 102 participating in a group conversation in which audio-video content streams and/or message content (e.g., text, images) are transmitted between the participant devices.

A virtual conference client **104** is able to communicate and exchange data with other virtual conference clients **104** and with the virtual conference server system **108** via the network **112**. The data exchanged between virtual conference clients **104**, and between a virtual conference client **104** and the virtual conference server system **108**, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., video, audio, other multimedia data, text).

The virtual conference server system **108** provides server-side functionality via the network **112** to a particular virtual conference client **104**. For example, with respect to transmitting audio and/or video streams, the virtual conference client **104** (e.g., installed on a first client device **102**) may facilitate in transmitting streaming content to the virtual conference server system **108** for subsequent receipt by other participant devices (e.g., one or more second client devices **102**) running respective instances of the virtual conference client **104**.

The streaming content can correspond to audio and/or video content captured by sensors (e.g., microphones, video cameras) on the client devices **102**, for example, corresponding to real-time video and/or audio capture of the users (e.g., faces) and/or other sights and sounds captured by the respective device. The streaming content may be supplemented with other audio/visual data (e.g., animations, overlays, emoticons and the like) and/or message content (e.g., text, stickers, emojis, other image/video data), for example, in conjunction with extension applications and/or widgets associated with the virtual conference client **104**.

While certain functions of the virtual conferencing system **100** are described herein as being performed by either a virtual conference client **104** or by the virtual conference server system **108**, the location of certain functionality either within the virtual conference client **104** or the virtual conference server system **108** may be a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the virtual conference server system **108** but to later migrate this technology and functionality to the virtual conference client **104** where a client device **102** has sufficient processing capacity.

The virtual conference server system **108** supports various services and operations that are provided to the virtual conference client **104**. Such operations include transmitting data to, receiving data from, and processing data generated by the virtual conference client **104**. This data may include the above-mentioned streaming content and/or message content, client device information, and social network information, as examples. Data exchanges within the virtual conferencing system **100** are invoked and controlled through functions available via user interfaces (UIs) of the virtual conference client **104**.

Turning now specifically to the virtual conference server system **108**, an Application Program Interface (API) server **114** is coupled to, and provides a programmatic interface to, application servers **118**. The application servers **118** are communicatively coupled to a database server **124**, which facilitates access to a database **126** that stores data associated with virtual conference content processed by the application servers **118**. Similarly, a web server **116** is coupled to the application servers **118**, and provides web-based interfaces to the application servers **118**. To this end, the web server **116** processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

The Application Program Interface (API) server **114** receives and transmits virtual conference data (e.g., com-

mands, audio/video payloads) between the client device **102** and the application servers **118**. Specifically, the Application Program Interface (API) server **114** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the virtual conference client **104** in order to invoke functionality of the application servers **118**. The Application Program Interface (API) server **114** exposes various functions supported by the application servers **118**, including account registration, login functionality, the streaming of audio and/or video content, and/or the sending and retrieval of message content, via the application servers **118**, from a particular virtual conference client **104** to another virtual conference client **104**, the retrieval of a list of contacts of a user of a client device **102**, the addition and deletion of users (e.g., contacts) to a user graph (e.g., a social graph), and opening an application event (e.g., relating to the virtual conference client **104**).

The application servers **118** host a number of server applications and subsystems, including for example a virtual conference server **120** and a social network server **122**. The virtual conference server **120** implements a number of virtual conference processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., streaming content) included in audio-video feeds received from multiple instances of the virtual conference client **104**. Other processor and memory intensive processing of data may also be performed server-side by the virtual conference server **120**, in view of the hardware requirements for such processing.

The social network server **122** supports various social networking functions and services and makes these functions and services available to the virtual conference server **120**. To this end, the social network server **122** maintains and accesses a user graph **304** (as shown in FIG. 3) within the database **126**. Examples of functions and services supported by the social network server **122** include the identification of other users of the virtual conferencing system **100** with which a particular user has relationships (e.g., contacts such as friends, colleagues, teachers, students, and the like).

In one or more embodiments, a user interacting via the virtual conference client **104** running on a first client device **102** may select and invite participant(s) to a virtual conference. For example, the participants may be selected from contacts maintained by the social network server **122**. In another example, the participants may be selected from contacts included within a contact address book stored in association with the first client device **102** (e.g., in local memory or in a cloud-based user account). In another example, the participants may be selected by the user manually entering email addresses and/or phone numbers of the participants.

The user at the first client device **102** may initiate the virtual conference by selecting an appropriate user interface element provided by the virtual conference client **104**, thereby prompting the invited participants, at their respective devices (e.g., one or more second client devices **102**), to accept or decline participation in the virtual conference. When the participant(s) have accepted the invitation (e.g., via the prompt), the virtual conference server system **108** may perform an initialization procedure in which session information is published between the participant client devices **102**, including the user who provided the invite. Each of the participant client devices **102** may provide respective session information to the virtual conference server system **108**, which in turn publishes the session information to the other participant client devices **102**. The session information for each client device **102** may include

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content stream(s) and/or message content that is made available by the client device **102**, together with respective identifiers for the content stream(s) and/or message content.

As described below with respect to FIG. **2**, the virtual conference may correspond to a virtual space which includes one or more rooms (e.g., virtual rooms). The virtual space and its corresponding rooms may have been created at least in part by the inviting user and/or by other users. In this manner, an end user may act as an administrator, who creates their own virtual spaces with rooms, and/or designs a virtual space based on preset available rooms.

FIG. **2** is a block diagram illustrating further details regarding the virtual conferencing system **100**, according to some examples. Specifically, the virtual conferencing system **100** is shown to comprise the virtual conference client **104** and the application servers **118**. The virtual conferencing system **100** embodies a number of subsystems, which are supported on the client-side by the virtual conference client **104** and on the server-side by the application servers **118**. These subsystems include, for example, a virtual space creation system **202** which implements a virtual space design interface **204**, and a virtual space participation system **206** which implements a virtual space navigation interface **208**.

The virtual space creation system **202** provides for a user to design one or more virtual space(s) in which participants may engage in virtual conferencing. In one or more embodiments, a virtual space corresponds to an environment with one or more rooms configured to accommodate virtual conferencing.

The virtual space may be created and/or selected (e.g., from among a set of predefined virtual spaces with rooms) by an end user who wishes to invite other users for virtual conferencing. In addition, the individual rooms of a virtual space may be newly-created and/or selected (e.g., from among a set of predefined rooms) by the end user. In one or more embodiments, the virtual space creation system **202** includes a virtual space design interface **204**, which is usable by the end user to design a virtual space, including creating and/or selecting rooms for including in the virtual space.

As discussed below with respect to FIG. **4**, the virtual space design interface **204** enables an end user (e.g., acting as an administrator) to select and/or position multiple elements within in a room. Examples of elements include, but are not limited to, participant video elements (e.g., for displaying the respective video feeds of participants), chat interfaces (e.g., for participants to provide text-based messages, stickers and/or reactions within a room), breakout buttons (e.g., for shuffling from a first room to one or more second rooms), and/or other user-definable elements for performing certain actions (e.g., speaking into a virtual microphone, querying an administrator via a button, and the like).

The virtual space participation system **206** is configured to perform virtual conferencing among participants within a virtual space. The participants may include the end user (e.g., administrator) who created the virtual space, as well as those users who were invited to participate in virtual conferencing with respect to the virtual space created/selected by the end user. The virtual space participation system **206** includes a virtual space navigation interface **208** (e.g., discussed below with respect to FIG. **5**) that allows participants to navigate between the rooms of a virtual space, and to participate in virtual conferencing with respect to the rooms.

In one or more embodiments, the virtual space creation system **202** and the virtual space participation system **206**

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provide for an end user (e.g., an administrator) to create different types of environments (e.g., virtual spaces with rooms) for virtual conferencing, and for participants to engage in virtual conferencing within such environments. Examples of such virtual conferencing include, but are not limited to: business meetings, seminars, presentations, classroom lectures, teacher office hours, concerts, reunions, virtual dinners, escape rooms, and the like.

FIG. **3** is a schematic diagram illustrating data structures **300**, which may be stored in the database **126** of the virtual conference server system **108**, according to certain examples. While the content of the database **126** is shown to comprise a number of tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

The database **126** includes profile data **302**, a user graph **304** and a user table **306** relating to the users (participants) of the virtual conferencing system **100**. The user table **306** stores user data, and is linked (e.g., referentially) to the user graph **304** and the profile data **302**. Each user of the virtual conferencing system **100** is associated with a unique identifier (email address, telephone number, social network identifier, etc.).

The user graph **304** stores (e.g., in conjunction with the social network server **122**) information regarding relationships and associations between users. Such relationships may be social, professional (e.g., work at a common corporation or organization) interested-based or activity-based, merely for example. As noted above, the user graph **304** may be maintained and accessed at least in part by the social network server **122**.

The profile data **302** stores multiple types of profile data about a particular user. The profile data **302** may be selectively used and presented to other users of the virtual conferencing system **100**, based on privacy settings specified by a particular user. The profile data **302** includes, for example, a user name, telephone number, email address, and/or settings (e.g., notification and privacy settings), as well as a user-selected avatar representation.

The database **126** further includes a virtual spaces table **308**. As noted above, a virtual space corresponds to an environment with one or more rooms configured to accommodate virtual conferencing. A virtual space may be newly-created by a user, or may be included within one or more sets of public virtual spaces made available (e.g., by other users, system administrators, and the like) for virtual conferencing. The virtual spaces table **308** stores information representing the one or more sets of public virtual spaces, as well as any private virtual space(s) created by a user (e.g., in a case where the particular user did not make such virtual space(s) public).

In one or more embodiments, the virtual spaces table **308** stores associations between its virtual spaces and users (e.g., within the user table **306**) who selected those virtual spaces. In this manner, it is possible for a particular user to have one or more virtual spaces associated therewith. Moreover, the database **126** includes a rooms table **310** which may be associated with the virtual spaces within the virtual spaces table **308**. As noted above, a room may be newly-created by a user, or may be included within one or more sets (e.g., galleries) of public rooms made available for user selection. The rooms table **310** stores information representing the one or more sets of rooms, as well as any private room(s) created by the user (e.g., in a case where the particular user did not make such room(s) public). The stored information is usable by the virtual conferencing system **100** to create the corresponding rooms for use in a virtual space. In one or more

embodiments, the stored information may further include recordings (e.g., audio and/or video recordings) of a particular virtual conference, for subsequent playback by corresponding participants.

FIG. 4 illustrates a virtual space design interface **204** with interface elements for designing a virtual space, in accordance with some example embodiments. Designing the virtual space may include creation and/or selection of rooms for including in the virtual space. The virtual space design interface **204** includes a menu interface **402**, a room elements interface **404**, an element properties interface **406**, a controls interface **408**, a room list interface **410**, a room canvas interface **412**, and an administrator name **414**. It is noted that elements **402-414** correspond to an example of interface elements for the virtual space design interface **204**, and that additional, fewer and/or different interface elements may be used.

An administrator (e.g., corresponding to administrator name **414**) may use the various interface elements to design a virtual space. In one or more embodiments, the menu interface **402** includes user-selectable categories (e.g., menu headings) relating to a virtual space (e.g., “workspace”), rooms within the virtual space, and/or elements within a room. For example, the workspace category is user-selectable for presenting options (e.g., via a drop-down list) to manage settings for the virtual space, manage invites for the virtual space, manage versions of a virtual space, publish the virtual space (e.g., for future use by users), manage virtual space publications, and/or to start/manage recordings (e.g., audio and/or video recordings) with respect to the virtual space.

The room category of the menu interface **402** is user-selectable for presenting options (e.g., via a drop-down list) to manage settings for a room within the virtual space, set a room background, set an order for the rooms listed in the room list interface **410**, create a new room, import a room from a set of available rooms, remove a room, publish a room, manage room publications, and/or to start/manage recordings with respect to a room.

In addition, the element category is user-selectable for presenting options (e.g., via a drop-down list) to insert elements into a room, insert shapes into a room, foreground/background elements, arrange/position elements, and/or group elements. Examples of elements include, but are not limited to: an action button, analog clock, audience question board, backpack item, breakout button, chat, closed caption display, closed caption input, countdown, clock, digital clock, doorbell, double-sided image, feedback, image, multiuser video chat, music, participant audio mixer, participant count, participant video, picture strip, poll, random source, room preview, scheduled time, sound effect, stopwatch, take picture, text, timer, user search, video, waiting list, web media, website. Examples of shapes include, but are not limited to, a circle, rectangle and triangle.

The users category of the menu interface **402** is user-selectable for presenting options (e.g., via a drop-down list) to manage users/participants of the virtual space (e.g., adding tags for participants, so as to distinguish between roles such as an administrator or an attendee/participant). In addition, the edit category is user-selectable for performing edit operations (e.g., undo, redo, cut, copy, paste), and the help category is user-selectable for performing help operations (e.g., getting started, discord, live help, submitting feedback).

In one or more embodiments, the room elements interface **404** includes user-selectable icons for inserting elements (e.g., corresponding to a subset of those available via the

above-mentioned element category) into a current room. For example, the elements may be added and/or positioned within the current room by selecting the element and dragging the selected element onto the room canvas interface **412**, which represents the layout of the current room.

In one or more embodiments, the room elements interface **404** include icons including but not limited to: a text icon for adding text to a room; a participant video icon for adding a single participant video element (e.g., an interface element which is selectable by a single participant for displaying that participant’s video feed) to a room; a multiuser video icon for adding a multiple participant video element (e.g., an interface element which is selectable by one or more participants for displaying the video feeds for those participants) to a room; a chat icon for adding a chat interface (e.g., for messaging using text, stickers, emojis, etc.) to a room; a video playback icon for adding a video playback element (e.g., screen) to a room for playback of a selected video; a background icon for selecting a background color/gradient, image or video to a room; an action icon for adding an action element (e.g., button) to a room for performing a user-defined action (e.g., speaking into a virtual microphone, querying an administrator via a button, and the like); and/or a breakout button for adding a breakout element (e.g., button) for shuffling selected participants between the current room and one or more other rooms.

In one or more embodiments, the element properties interface **406** include various fields for setting configuration properties for above-described room elements. For example, with respect to elements in general (e.g., text, single participant video element, multi participant video element, chat interface, video element, background image, action element, breakout button), the element properties interface **406** includes fields for setting the element title/name, opacity, gradient, style, layout, borders/corners, shadows, interaction (e.g., to what extent participant(s) may delete, modify, resize the element), filtering, full screen status, conditions, accessibility and actions for the element. For the single participant video element, the element properties interface **406** includes further fields for setting the manner in which users are placed into the single participant video element during virtual conferencing (e.g., automatically, manually by the participant and/or the administrator end user). In addition, for the chat interface, the element properties interface **406** includes further properties for setting who (e.g., administrator and/or participants) can provide chat input, and/or which types of input (e.g., text, stickers, emojis, etc.) are available. For the action element, the element properties interface **406** includes further properties for setting what type of action is to be performed in response to user selection of the action element (e.g., button). Moreover, for the breakout element, the element properties interface **406** includes further properties for selecting participants and/or breakout rooms.

In one or more embodiments, the element properties interface **406** includes fields for setting configuration properties for the room canvas interface **412**. For example, the element properties interface **406** includes fields selecting a number of fake participants (e.g., simulated video feeds) in order to visualize multiple users, selecting music (e.g., background music), and/or selecting which reaction buttons are available for participants to indicate real-time reactions with respect to virtual conferencing within a room.

In one or more embodiments, the controls interface **408** includes user-selectable icons corresponding to controls (e.g., administrative controls) for the virtual space. For example, the controls interface **408** include icons including but not limited to: a director mode icon for toggling between

a director mode for designing a room and a user mode for viewing the room within the virtual space design interface **204** (e.g., with the director mode including the room elements interface **404** and the element properties interface **406** while the user mode does not); a view icon for viewing the room within the virtual space navigation interface **208**; a share screen icon (e.g., for collaborative design with other user(s) such as co-administrators); a microphone icon for enabling or disabling the microphone; a help icon (e.g., getting started, discord, live help, submitting feedback); an invite icon (e.g., for displaying an invite link for sending to participants to visit the virtual space); a settings icon (e.g., for selecting the end user's video and audio devices for the virtual conferencing, and for selecting a user avatar); and/or an exit icon for exiting the virtual space design interface **204**.

In one or more embodiments, the room list interface **410** displays the list of rooms for the virtual space. Each listed room is user selectable to switch to edit (e.g., in director mode) and/or view (e.g., in user mode) the selected room. As noted above, the list of rooms may be modified (e.g., by adding, importing and/or removing rooms) via the options within the room category of the menu interface **402**.

FIG. 5 illustrates a virtual space navigation interface **208** with interface elements to navigate between the rooms of a virtual space and to participate in virtual conferencing with respect to the rooms, in accordance with some example embodiments. The virtual space navigation interface **208** includes a controls interface **502**, a room list interface **504**, a current room interface **506**, a participant video element **508** and a participant video element **510**. It is noted that elements **502-512** correspond to an example of interface elements for the virtual space navigation interface **208**, and that additional, fewer and/or different interface elements may be used.

In one or more embodiments, the controls interface **502** includes user-selectable icons corresponding to controls (e.g., administrative controls) for the virtual space. For example, the controls interface **408** include icons including but not limited to: an edit icon for redirecting to the virtual space design interface **204** to edit the current room; a volume icon for adjusting a volume level for the current room; a share screen icon (e.g., for allowing others to view the room without necessarily joining the room); a microphone icon for muting and unmuting the microphone; a help icon (e.g., getting started, discord, live help, submitting feedback); an invite icon (e.g., for displaying an invite link for participants to visit the virtual space); a settings icon (e.g., for selecting the end user's video and audio devices for the virtual conferencing, and for selecting a user avatar); and/or an exit icon for exiting the virtual space design interface **204**.

In one or more embodiments, the room list interface **504** displays the list of rooms for the virtual space. Each listed room is user selectable to switch to the selected room (e.g., for virtual conferencing). The selected room is presented as a current room within the current room interface **506**. In this manner, a participant may navigate among the multiple rooms available within the virtual space. Alternatively or in addition, navigation between rooms is possible via a virtual space map interface (not shown) which depicts a map view of the virtual space (e.g., a floor plan) and its corresponding rooms, with each room being user selectable to navigate thereto. Alternatively or in addition, navigation between rooms is further possible by positioning a navigation button (not shown) within a room, where user selection of the button results in navigating to another room (e.g., a predefined room). As noted above, the virtual space design

interface **204** allows for the design of a virtual space and its corresponding rooms. As such, navigation between rooms is based at least in part on the design of the virtual space (e.g., a virtual space may include one or more of the above-mentioned room list interface **504**, the virtual space map/floor plan interface and/or the navigation button).

With respect to the current room interface **506**, each participant is represented as a respective participant video element. As noted above, a participant video element corresponds to an interface element (e.g., a box) which is selectable by a single participant for displaying that participant's video feed. The example of FIG. 5 includes a first participant associated with the participant video element **508** and a second participant associated with the participant video element **510**. In one or more embodiments, with respect to the perspective of the first participant, the participant video element **510** showing the feed of the second participant may include participant button(s) **512**. For example, the participant button(s) **512** are selectable by the first participant so as to perform a predefined action (e.g., initiate a side conversation, designate the second participant to follow the first participant when the first participant moves rooms) with respect to the second participant.

While the example of FIG. 5 illustrates two participants, it is possible for the current room interface **506** to accommodate additional participants for virtual conferencing. The additional participants may be positioned (e.g., automatically and/or manually by dragging) based on the positioning of participant video elements as designed by the virtual space design interface **204**.

In one or more embodiments, the virtual space navigation interface **208** may vary based on whether a given participant is an administrator or another participant (e.g., an attendee). For example, some participant video elements may be designated (e.g., via the virtual space design interface **204**) for administrators, while other participant video elements are designated for other participants. The virtual conference server system **108** is configured to distinguish between these administrator or other participant roles, for example, based on the above-described tags assigned to participants via the users category of the menu interface **402** provided by the virtual space design interface **204**.

FIG. 6 is an interaction diagram illustrating a process **600** for providing a side conversation within a virtual conferencing system, in accordance with some example embodiments. For explanatory purposes, the process **600** is described herein with reference to a first client device **602**, a second client device **604**, one or more third client device(s) **606**, and the virtual conference server system **108**. Each of the first client device **602**, the second client device **604** and the third client device(s) **606** may correspond to a respective client device **102**. The process **600** is not limited to the first client device **602**, the second client device **604**, the third client device(s) **606** and the virtual conference server system **108**. Moreover, one or more blocks (or operations) of the process **600** may be performed by one or more other components of the first client device **602**, the second client device **604**, the third client device(s) **606** or the virtual conference server system **108**, and/or by other suitable devices. Further for explanatory purposes, the blocks (or operations) of the process **600** are described herein as occurring in serial, or linearly. However, multiple blocks (or operations) of the process **600** may occur in parallel or concurrently. In addition, the blocks (or operations) of the process **600** need not be performed in the order shown and/or one or more blocks (or operations) of the process **600** need not be performed and/or can be replaced by other

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operations. The process 600 may be terminated when its operations are completed. In addition, the process 600 may correspond to a method, a procedure, an algorithm, etc.

Each of the first client device 602, the second client device 604 and the third client device(s) 606 have instances of the virtual conference client 104 installed thereon. In the example of FIG. 6, the first client device 602, the second client device 604, and the third client device(s) 606 are associated with a respective first participant, second participant and one or more third participant(s) of the virtual conference server system 108. For example, the first participant may be associated with a first user account of the virtual conference server system 108, the second participant may be associated with a second user account of the virtual conference server system 108, and the third participant(s) may be associated with third user account(s) of the virtual conference server system 108.

As noted above, the first, second and third participants are identifiable by the virtual conference server system 108 based on unique identifiers (e.g., email addresses, telephone numbers) associated with respective user accounts for the first, second and third participants. In one or more embodiments, the virtual conference server system 108 implements and/or works in conjunction with a social network server 122 which is configured to identify contacts with which a particular user has relationships. For example, the first, second and third participants may be contacts with respect to the virtual conference server system 108.

As described herein, the virtual conferencing system 100 provides for the first participant to initiate a side conversation with the second participant, without disrupting other participants (e.g., third participant(s)). At operations 608-612, the virtual conference server system 108 provides for real-time communication between the participant devices, namely the first client device 602, the second client device 604 and the third client device(s) 606. For example, virtual conference server system 108 provides for receiving and transmitting of data between the devices 602-606, including one or more of audio, video, image, text and/or other signals. As noted above, the real-time communication (e.g., virtual conferencing) may occur within the context of a current room within a virtual space.

In one or more embodiments, audio (e.g., an audio stream) is transmitted via a first audio channel. For example, the first audio channel corresponds to a primary communication stream between the plural participants. The communication may be one way (e.g., from a presenter to the remaining participants) or multi-way (e.g., between two or more of the participants), based on the configuration/design of the room (e.g., as set via the element properties interface 406).

In one or more embodiments, each participant is represented by a respective participant video element (e.g., elements 508-510 in FIG. 5). As noted above, from the perspective of the first participant, the first client device 602 is configured to display a participant button for starting a side conversation with the second participant (e.g., a participant button(s) 512 displayed within the participant video element of the second participant) or with the third participant (e.g., a participant button(s) 512 displayed within the participant video element of the third participant). As discussed below with respect to FIGS. 7A-7C, the participant button for initiating the side conversation is depicted as an ear icon.

At block 614, the first client device 602 receives user selection (e.g., by the first participant) of the participant button to initiate a side conversation with the second par-

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ticipant. The first client device 602 sends an indication of the user selection to the virtual conference server system 108 (operation 616).

In response, the virtual conference server system 108 initiates one-way communication via a second audio channel between the first client device 602 and the second client device 604 (operation 618). The second audio channel is separate from the above-described first audio channel, for sending audio (e.g., an audio stream) that is input at the first client device 602 (e.g., via a microphone) to the second client device 604 for output to the second participant (e.g., via a speaker). As such, the virtual conference server system 108 provides for transmitting the audio stream from the first client device 602 to the second client device 604, while refraining from transmitting the audio stream to the third client device(s) 606.

In one or more embodiments, the second audio channel is initially configured for one-way communication from the first client device 602 to the second client device 604, such that the first participant can communicate to the second participant at the second client device 604, but the second participant cannot yet communicate back to the first participant at the first client device 602. In addition to providing the one-way communication, the virtual conference server system 108 at operation 618 may further prompt the second participant for two-way communication via the second audio channel.

For example, the virtual conference server system 108 may provide for a notification to be displayed on the second client device 604. The notification may indicate, to the second participant, that the first participant initiated a side conversation by which the first participant communicates with the second participant and not with the third participant(s). The notification may further indicate that the side conversation is one-way. The notification may further include a prompt, which is selectable by the second user to initiate two-way communication, to enable the second participant to communicate back to the first participant at the first client device 602.

At operation 620, the virtual conference server system 108 provides an indication of the side conversation to the third client device(s) 606. In one or more embodiments, the indication of the side conversation causes the third client device(s) 606 to display the participant button (e.g., ear icon) for the participant video element of the first participant to blink. For example, the blinking may be synchronized to pulsate with the audio stream corresponding to the one-way communication from the first client device 602 to the second client device 604 (e.g., such that the ear icon pulsates with the audio stream reaching certain audio levels). The blinking may indicate, to the third participant(s) that the first participant is involved in a side conversation with another participant, without necessarily identifying the other participant (i.e., the second participant).

At operation block 622, the second client device 604 receives user selection of the prompt for initiating two-way communication. The second client device 604 sends an indication of the user selection to the virtual conference server system 108 (operation 624). In response, the virtual conference server system 108 provides for updating the one-way communication to two-way communication via the second audio channel (operations 626-628). As noted above, the two-way communication provides for the first participant to communicate to the second participant at the second client device 604, and the second participant to communicate back to the first participant at the first client device 602.

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In addition, the virtual conference server system **108** provides an updated indication of the side conversation to the third client device(s) **606** (operation **630**). For example, the above-mentioned blinking is updated to synchronize with the audio stream corresponding to the two-way communication between the first client device **602** and the second client device **604** (e.g., such that the ear icon pulsates based on the audio streams from both the first client device **602** and from the second client device **604**). The blinking may indicate, to the third participant(s) that the first participant is involved in a side conversation with another participant, without necessarily identifying the other participant (i.e., the second participant).

In one or more embodiments, during the side conversation (e.g., during one-way communication and/or two-way communication via the second audio channel), the first client device **602** may update display of the participant video element(s) for the third participant(s). For example, the virtual conference client **104** running on the first client device **602** (e.g., in conjunction with the virtual conference server system **108**) may gray out, blur, or otherwise obfuscate display of the participant video element(s) for the third participant(s), while not obfuscating display of the participant video element for the second participant. In this manner, the display of the current room (e.g., current room interface **506**) on the first client device **602** more clearly indicates the side conversation with the second participant.

Moreover, during the side conversation, the second client device **604** may update display of the participant video element for the first participant. For example, the virtual conference client **104** running on the second client device **604** (e.g., in conjunction with the virtual conference server system **108**) may highlight, brighten, or otherwise emphasize display of the participant video element for the first participant, while not emphasizing display of the participant video element(s) for the third participant(s).

In one or more embodiments, the virtual conference server system **108** provides for reducing the audio level of the first audio channel (e.g., by a predefined percentage) for the second client device **604** during the one-way communication and/or the two-way communication, for example, to prioritize the audio stream on the second audio channel. Moreover, the virtual conference server system **108** may provide for reducing the audio level of the first audio channel (e.g., by a predefined percentage) for the first client device **602** during the two-way communication, for example, to emphasize the audio stream on the second audio channel.

Thus, the virtual conferencing system **100** as described herein allows participants to initiate and/or engage in a side conversation, without disrupting other participants during virtual conferencing. The interface elements for initiating and/or engaging in the side conversation are intuitive and simple. Without such interface elements, the participants may otherwise be required to separately message each other to a separate conversation (e.g., in another room). The virtual conferencing system **100** reduces such additional efforts by the participants, thereby saving time for the participants, and reducing computational resources/processing power for the virtual conferencing system **100**.

FIGS. 7A-7C illustrate various interface elements relating to providing a side conversation between participants of a virtual conference, in accordance with some example embodiments. As noted above, a participant video element (e.g., participant video element **510**) may include one or more participant button(s) (e.g., participant button(s) **512**). FIG. 7A depicts one example, in which a side conversation

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button **702** depicted as an ear is user-selectable to initiate a side conversation with another participant.

As described above with respect to FIG. 6, in response to user selection (e.g., by a first participant) of the side conversation button **702** (e.g., presented within the participant video element of a second participant), the virtual conference server system **108** provides for initiating a one-way communication between the first participant and the second participant via a second audio channel. The virtual conference server system **108** may provide a notification **704** to the first participant. The notification confirms, to the first participant, the one-way communication from the first participant to the second participant, while refraining from transmitting the corresponding audio stream to the third participant(s).

The notification **704** indicates, to the second participant, that the first participant initiated a side conversation by which the first participant communicates with the second participant and not with third participant(s). In one or more embodiments, the notification **704** includes a button **706** which is selectable by the first participant to exit the side conversation (e.g., thereby terminating communication via the second audio channel).

With respect to FIG. 7B, the virtual conference server system **108** further provides a notification **708** to the second participant as described above with respect to FIG. 6. The notification **708** indicates, to the second participant, that the first participant initiated a side conversation. The notification **708** further indicates that the side conversation is one-way, and includes a button **710**, which is selectable by the second user to initiate two-way communication, to enable the second participant to communicate back to the first participant at the first client device **602**.

Moreover, regarding FIG. 7C, the virtual conference server system **108** provides a notification **712** to the second participant, confirming the two-way communication between the first and second participants. The notification **712** includes a button **714** which is selectable by the second participant to exit the side conversation (e.g., thereby terminating two-way communication via the second audio channel).

FIG. 8 is a flowchart illustrating a process **800** for participating in a side conversation within a virtual conferencing system, in accordance with some example embodiments. For explanatory purposes, the process **800** is primarily described herein with reference to the first client device **602**, the second client device **604**, the third client device(s) **606** and the virtual conference server system **108** of FIG. 1 and FIG. 2. However, one or more blocks (or operations) of the process **800** may be performed by one or more other components, and/or by other suitable devices. Further for explanatory purposes, the blocks (or operations) of the process **800** are described herein as occurring in serial, or linearly. However, multiple blocks (or operations) of the process **800** may occur in parallel or concurrently. In addition, the blocks (or operations) of the process **800** need not be performed in the order shown and/or one or more blocks (or operations) of the process **800** need not be performed and/or can be replaced by other operations. The process **800** may be terminated when its operations are completed. In addition, the process **800** may correspond to a method, a procedure, an algorithm, etc.

The virtual conference server system **108** provides, in association with a virtual conference among plural participants, a first audio channel for sharing among the plural

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participants (block 802). The first audio channel may correspond to multi-way communication between the plural participants.

The virtual conference server system 108 provides, for each of the plural participants, display of a participant video element which corresponds to the participant and which includes a user-selectable button to initiate a side conversation with the participant (block 804). For each of the plural participants, the participant video element may include a video feed for the participant.

The virtual conference server system 108 receives indication of a user selection, by a first participant of the plural participants, of the user-selectable button for initiating a side conversation with a second participant of the plural participants (block 806). The virtual conference server system 108 provides, in response to receiving the indication, a second audio channel from a first client device 602 associated with the first participant to a second client device 604 associated with the second participant, in conjunction with providing the first audio channel to the second client device 604 (block 808). The virtual conference server system 108 may reduce an audio level of the first audio channel to the second device during audio transmission via the second audio channel to the second device.

The virtual conference server system 108 may provide, in response to receiving the indication, display of a notification on the second client device 604, the notification indicating that the first participant initiated one-way communication with the second participant. The virtual conference server system 108 may provide, together with the notification, display of a second user-selectable element on the second client device 604, the second user-selectable element for initiating two-way communication between the first client device 602 and the second client device 604, receive indication of a second user selection, by the second participant, of the second user-selectable element, and provide, in response to receiving the indication of the second user selection, for updating the second audio channel from one-way communication to two-way communication between the first client device 602 and the second client device 604. The virtual conference server system 108 may provide, in response to receiving the indication of the second user selection, display of a second notification on the second client device 604, the second notification indicating that the first participant and the second participant are participating in two-way communication.

The virtual conference server system 108 may refrain from providing the second audio channel to remaining participants of the plural participants. The virtual conference server system 108 may cause, on respective third client device(s) 606 associated with the remaining participants, display of the user-selectable button for the first participant to pulsate in synchronization with audio levels of the second audio channel (e.g., based on either one-way or two-way communication between the first client device 602 and the second client device 604). The virtual conference server system 108 may cause, on the first client device 602, display of the participant video elements corresponding to the remaining participants to be different than display of the participant video element corresponding to the second participant. The virtual conference server system 108 may cause, on the second client device 604, display of the participant video element corresponding to the first participant to be different than display of the participant video elements corresponding to the remaining participants.

FIG. 9 is a diagrammatic representation of the machine 900 within which instructions 910 (e.g., software, a pro-

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gram, an application, an applet, an app, or other executable code) for causing the machine 900 to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions 910 may cause the machine 900 to execute any one or more of the methods described herein. The instructions 910 transform the general, non-programmed machine 900 into a particular machine 900 programmed to carry out the described and illustrated functions in the manner described. The machine 900 may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine 900 may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine 900 may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions 910, sequentially or otherwise, that specify actions to be taken by the machine 900. Further, while only a single machine 900 is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions 910 to perform any one or more of the methodologies discussed herein. The machine 900, for example, may comprise the client device 102 or any one of a number of server devices forming part of the virtual conference server system 108. In some examples, the machine 900 may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

The machine 900 may include processors 904, memory 906, and input/output I/O components 902, which may be configured to communicate with each other via a bus 940. In an example, the processors 904 (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor 908 and a processor 912 that execute the instructions 910. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. 9 shows multiple processors 904, the machine 900 may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

The memory 906 includes a main memory 914, a static memory 916, and a storage unit 918, both accessible to the processors 904 via the bus 940. The main memory 906, the static memory 916, and storage unit 918 store the instructions 910 embodying any one or more of the methodologies or functions described herein. The instructions 910 may also reside, completely or partially, within the main memory 914, within the static memory 916, within machine-readable medium 920 within the storage unit 918, within at least one

of the processors **904** (e.g., within the Processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **900**.

The I/O components **902** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **902** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **902** may include many other components that are not shown in FIG. 9. In various examples, the I/O components **902** may include user output components **926** and user input components **928**. The user output components **926** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input components **928** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

In further examples, the I/O components **902** may include biometric components **930**, motion components **932**, environmental components **934**, or position components **936**, among a wide array of other components. For example, the biometric components **930** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **932** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

The environmental components **934** include, for example, one or more cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

With respect to cameras, the client device **102** may have a camera system comprising, for example, front cameras on a front surface of the client device **102** and rear cameras on a rear surface of the client device **102**. The front cameras may, for example, be used to capture still images and video

of a user of the client device **102** (e.g., "selfies"), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the client device **102** may also include a 360° camera for capturing 360° photographs and videos.

Further, the camera system of a client device **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the client device **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera and a depth sensor, for example.

The position components **936** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

Communication may be implemented using a wide variety of technologies. The I/O components **902** further include communication components **938** operable to couple the machine **900** to a network **922** or devices **924** via respective coupling or connections. For example, the communication components **938** may include a network interface Component or another suitable device to interface with the network **922**. In further examples, the communication components **938** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **924** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

Moreover, the communication components **938** may detect identifiers or include components operable to detect identifiers. For example, the communication components **938** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **938**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

The various memories (e.g., main memory **914**, static memory **916**, and memory of the processors **904**) and storage unit **918** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **910**), when executed by processors **904**, cause various operations to implement the disclosed examples.

The instructions **910** may be transmitted or received over the network **922**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **938**) and using any one of several well-known transfer protocols (e.g.,

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hypertext transfer protocol (HTTP)). Similarly, the instructions **910** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **924**.

FIG. **10** is a block diagram **1000** illustrating a software architecture **1004**, which can be installed on any one or more of the devices described herein. The software architecture **1004** is supported by hardware such as a machine **1002** that includes processors **1020**, memory **1026**, and I/O components **1038**. In this example, the software architecture **1004** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **1004** includes layers such as an operating system **1012**, libraries **1010**, frameworks **1008**, and applications **1006**. Operationally, the applications **1006** invoke API calls **1050** through the software stack and receive messages **1052** in response to the API calls **1050**.

The operating system **1012** manages hardware resources and provides common services. The operating system **1012** includes, for example, a kernel **1014**, services **1016**, and drivers **1022**. The kernel **1014** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **1014** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionality. The services **1016** can provide other common services for the other software layers. The drivers **1022** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1022** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

The libraries **1010** provide a common low-level infrastructure used by the applications **1006**. The libraries **1010** can include system libraries **1018** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **1010** can include API libraries **1024** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **1010** can also include a wide variety of other libraries **1028** to provide many other APIs to the applications **1006**.

The frameworks **1008** provide a common high-level infrastructure that is used by the applications **1006**. For example, the frameworks **1008** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks **1008** can provide a broad spectrum of other APIs that can be used by the applications **1006**, some of which may be specific to a particular operating system or platform.

In an example, the applications **1006** may include a home application **1036**, a contacts application **1030**, a browser application **1032**, a book reader application **1034**, a location application **1042**, a media application **1044**, a messaging

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application **1046**, a game application **1048**, and a broad assortment of other applications such as a third-party application **1040**. The applications **1006** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **1006**, structured in a variety of manners, such as object-oriented programming languages (e.g., Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application **1040** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application **1040** can invoke the API calls **1050** provided by the operating system **1012** to facilitate functionality described herein.

Glossary

“Carrier signal” refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

“Client device” refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

“Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1×RTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

“Component” refers to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate

circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

“Computer-readable storage medium” refers to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium” and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure.

“Machine storage medium” refers to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and

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removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure. The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.”

“Non-transitory computer-readable storage medium” refers to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine.

“Signal medium” refers to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

What is claimed is:

1. A method, comprising:

providing, in association with a virtual conference among plural participants, a first audio channel for sharing among the plural participants;

providing, for each of the plural participants, display of a participant video element which corresponds to the participant and which includes a user-selectable button to initiate a side conversation with the participant;

receiving indication of a user selection, by a first participant of the plural participants, of the user-selectable button for initiating a side conversation with a second participant of the plural participants;

providing, in response to receiving the indication, a second audio channel from a first device associated with the first participant to a second device associated with the second participant, in conjunction with providing the first audio channel to the second device; and

providing, on respective devices associated with remaining participants of the plural participants, a notification of the side conversation between the first participant and the second participant, the notification being included within the participant video element of the first participant,

wherein the notification does not identify the second participant to the remaining participants.

2. The method of claim 1, further comprising:

providing, in response to receiving the indication, display of a notification on the second device, the notification indicating that the first participant initiated one-way communication with the second participant.

3. The method of claim 2, further comprising:

providing, together with the notification, display of a second user-selectable element on the second device, the second user-selectable element for initiating two-way communication between the first device and the second device;

receiving indication of a second user selection, by the second participant, of the second user-selectable element; and

providing, in response to receiving the indication of the second user selection, for updating the second audio

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channel from one-way communication to two-way communication between the first device and the second device.

4. The method of claim 3, further comprising:

providing, in response to receiving the indication of the second user selection, display of a second notification on the second device, the second notification indicating that the first participant and the second participant are participating in two-way communication.

5. The method of claim 1, further comprising:

causing, on the first device, display of the participant video elements corresponding to the remaining participants to be different than display of the participant video element corresponding to the second participant.

6. The method of claim 1, further comprising:

causing, on the second device, display of the participant video element corresponding to the first participant to be different than display of the participant video elements corresponding to the remaining participants.

7. The method of claim 1, wherein the first audio channel corresponds to multi-way communication between the plural participants.

8. The method of claim 7, further comprising:

reducing an audio level of the multi-way communication via the first audio channel to the second device during audio transmission via the second audio channel to the second device.

9. The method of claim 1, wherein for each of the plural participants, the participant video element includes a video feed for the participant.

10. A system comprising:

a processor; and

a memory storing instructions that, when executed by the processor, configure the processor to perform operations comprising:

providing, in association with a virtual conference among plural participants, a first audio channel for sharing among the plural participants;

providing, for each of the plural participants, display of a participant video element which corresponds to the participant and which includes a user-selectable button to initiate a side conversation with the participant;

receiving indication of a user selection, by a first participant of the plural participants, of the user-selectable button for initiating a side conversation with a second participant of the plural participants;

providing, in response to receiving the indication, a second audio channel from a first device associated with the first participant to a second device associated with the second participant, in conjunction with providing the first audio channel to the second device; and

providing, on respective devices associated with remaining participants of the plural participants, a notification of the side conversation between the first participant and the second participant, the notification being included within the participant video element of the first participant,

wherein the notification does not identify the second participant to the remaining participants.

11. The system of claim 10, the operations further comprising:

providing, in response to receiving the indication, display of a notification on the second device, the notification indicating that the first participant initiated one-way communication with the second participant.

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12. The system of claim 11, the operations further comprising:

providing, together with the notification, display of a second user-selectable element on the second device, the second user-selectable element for initiating two-way communication between the first device and the second device;

receiving indication of a second user selection, by the second participant, of the second user-selectable element; and

providing, in response to receiving the indication of the second user selection, for updating the second audio channel from one-way communication to two-way communication between the first device and the second device.

13. The system of claim 12, the operations further comprising:

providing, in response to receiving the indication of the second user selection, display of a second notification on the second device, the second notification indicating that the first participant and the second participant are participating in two-way communication.

14. The system of claim 12, the operations further comprising:

causing, on the first device, display of the participant video elements corresponding to the remaining participants to be different than display of the participant video element corresponding to the second participant.

15. The system of claim 12, the operations further comprising:

causing, on the second device, display of the participant video element corresponding to the first participant to be different than display of the participant video elements corresponding to the remaining participants.

16. The system of claim 10, wherein the first audio channel corresponds to multi-way communication between the plural participants.

17. The system of claim 16, the operations further comprising:

reducing an audio level of the multi-way communication via the first audio channel to the second device during

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audio transmission via the second audio channel to the second device.

18. The system of claim 10, wherein for each of the plural participants, the participant video element includes a video feed for the participant.

19. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a computer, cause the computer to perform operations comprising:

providing, in association with a virtual conference among plural participants, a first audio channel for sharing among the plural participants;

providing, for each of the plural participants, display of a participant video element which corresponds to the participant and which includes a user-selectable button to initiate a side conversation with the participant;

receiving indication of a user selection, by a first participant of the plural participants, of the user-selectable button for initiating a side conversation with a second participant of the plural participants;

providing, in response to receiving the indication, a second audio channel from a first device associated with the first participant to a second device associated with the second participant, in conjunction with providing the first audio channel to the second device; and

providing, on respective devices associated with remaining participants of the plural participants, a notification of the side conversation between the first participant and the second participant, the notification being included within the participant video element of the first participant,

wherein the notification does not identify the second participant to the remaining participants.

20. The non-transitory computer-readable storage medium of claim 19, the operations further comprising:

providing, in response to receiving the indication, display of a notification on the second device, the notification indicating that the first participant initiated one-way communication with the second participant.

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